Week 11: Monday, AST 5011: Astrophysical Systems

Stars & Stellar Populations

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Reference: Cimatti, Fraternali & Nipoti, Ch. 7

With material from Benedikt Diemer (UMD) and Frank van den Bosch (Yale).

The Initial Mass Function (IMF)

The initial mass function (IMF) describes the distribution of stellar masses at birth. It is one of the most fundamental quantities in astrophysics, as it determines:

- The ratio of massive to low-mass stars
- The total luminosity per unit mass formed
- The rate of supernova feedback
- The chemical enrichment of the ISM

The IMF is typically expressed as $dn/d\log_{10} m$, the number of stars per logarithmic mass interval. Several fitting functions are commonly used:

- Salpeter (1955): $\xi(m) \propto m^{-1.35}$ (single power law)
- Kroupa (2001): broken power law with shallower slope below $0.5\,M_\odot$
- Chabrier (2003): lognormal below $1\,M_\odot$, power law above

The key difference: Salpeter predicts more low-mass stars relative to high-mass stars compared to Kroupa/Chabrier. This affects mass-to-light ratios by a factor of ~ 1.5–2.

Demonstration: Cumulative IMF Distributions

The differential IMF $\xi(m)$ tells us the number of stars per logarithmic mass interval, but the cumulative distributions reveal where the total number and mass reside:

- $N(>m)/N_{\rm total}$: fraction of stars more massive than m. Most stars by number are low-mass (~ 0.1–$0.5\,M_\odot$).
- $M(>m)/M_{\rm total}$: fraction of total stellar mass in stars more massive than m. A significant fraction of the mass is in intermediate-mass stars (~ 1–$8\,M_\odot$), even though they are

outnumbered by low-mass stars.

The Salpeter IMF predicts more low-mass stars (steeper drop in cumulative number) compared to Kroupa/Chabrier, which have flatter low-mass slopes (adapted from CFN Chapter 7).

Exercise 1: IMF Mass Budget

For each IMF, compute:

1. The number fraction of stars with $m > 8\,M_\odot$ (massive stars that explode as core-collapse supernovae)
2. The mass fraction in massive stars ($m > 8\,M_\odot$)
3. The mean stellar mass $\langle m \rangle$

Use the mass range 0.08–$150\,M_\odot$. The integrals are:

$$f_{\rm number} = \frac{\int_8^{150} \xi(m)\,d\log m}{\int_{0.08}^{150} \xi(m)\,d\log m}, \quad f_{\rm mass} = \frac{\int_8^{150} m\,\xi(m)\,d\log m}{\int_{0.08}^{150} m\,\xi(m)\,d\log m}$$

Star Formation Histories

The star formation history (SFH) describes how the star formation rate of a galaxy evolves with time. Common parameterizations include:

1. Exponential decay: $\psi(t) \propto e^{-t/\tau}$ — rapid initial burst followed by decline
2. Delayed-τ model: $\psi(t) \propto (t/\tau)\,e^{-t/\tau}$ — rises to a peak then declines
3. Lognormal: peaked SFH with asymmetric tails

The parameter τ controls the timescale: small τ means a short burst (forming an “old, red” population), while large τ means extended star formation (a “young, blue” population).

Demonstration: Stellar Mass Assembly

The SFH determines when a galaxy assembles its stellar mass. The cumulative mass fraction $f_*(t) = \int_0^t \psi(t')\, dt' / \int_0^{t_{\rm age}} \psi(t')\, dt'$ shows how quickly mass builds up.

- Short τ (burst): mass is assembled early, producing an old, red population
- Long τ (extended SF): mass builds up gradually, resulting in a younger, bluer population
- Delayed-τ: mass assembly peaks later, with a more gradual onset

This also connects to "downsizing" — more massive galaxies tend to have shorter effective τ, forming their stars earlier despite residing in more massive halos (adapted from CFN Chapter 7).

Stellar Mass-to-Light Ratios

The key challenge in galaxy observations is converting luminosity (what we observe) to stellar mass (what we want to know). The conversion factor is the mass-to-light ratio M_*/L.

For a simple stellar population (SSP):

- Young populations are luminous and have low $M_*/L \sim 0.1$
- Old populations are faint per unit mass and have high $M_*/L \sim 3\text{–}5$

The M_*/L ratio correlates tightly with galaxy color because both are driven by the age of the stellar population:

- Blue galaxies (star-forming) → low M_*/L
- Red galaxies (quenched) → high M_*/L

Zibetti et al. (2009) found a useful empirical relation:

$$\log_{10}(M_*/L_r) = -0.840 + 1.654 \times (g - r)$$

This allows us to estimate M_* from just two observables: r-band luminosity and $g - r$ color.

Exercise 2: Estimating the Milky Way's Stellar Mass

Use the Zibetti et al. (2009) M_*/L–color relation to estimate the stellar mass of the Milky Way.

Given (Licquia & Newman 2016):

- $M_r = -20.97 + 5\log_{10}(h)$ (absolute r-band magnitude)
- $g - r = 0.678$
- $M_{r,\odot} = 4.68$ (solar absolute magnitude in r-band)

Steps:

1. Compute M_*/L_r from the color
2. Compute L_r from the absolute magnitude: $L_r = 10^{0.4(M_{r,\odot} - M_r)}$
3. Compute $M_* = (M_*/L_r) \times L_r$
4. Compare to the literature value: $M_* = 5.7 \times 10^{10}\, M_\odot$

The Star Formation Main Sequence

One of the most important observed correlations in galaxy evolution is the star formation main sequence (SFMS): a tight, nearly linear (in log-log) relation between stellar mass and star formation rate for star-forming galaxies.

$$\log_{10}(\mathrm{SFR}) \approx 0.8 \times \log_{10}(M_*) - 6.5$$

(approximately, at $z \sim 0$; Speagle et al. 2014)

Key features:

- Star-forming galaxies follow the main sequence
- Quenched galaxies lie below it (the "red and dead" cloud)
- The specific SFR (sSFR $= \mathrm{SFR}/M_*$) decreases with mass — more massive galaxies are less efficient at forming stars per unit mass
- There is a bimodality in sSFR: star-forming vs quenched populations

Exercise 3: The Star Formation Main Sequence from SDSS

Using the SDSS galaxy catalog:

1. Plot the **SFR–M_* relation (the star formation main sequence) as a 2D histogram
2. Overplot the Speagle et al. (2014) relation
3. Compute and plot the specific SFR** ($\mathrm{sSFR} = \mathrm{SFR}/M_*$) vs M_*
4. Plot the sSFR distribution to see the bimodality between star-forming and quenched galaxies

Demonstration: Color–sSFR Correlation

The bimodality in sSFR maps directly onto the color bimodality of galaxies. The figure below shows $g - r$ color vs. specific SFR for SDSS galaxies, revealing a remarkably tight correlation: blue galaxies are star-forming, red galaxies are quenched, and the boundary falls near $\mathrm{sSFR} \sim 10^{-11}\,\mathrm{yr}^{-1}$. This is the observational basis for using color as a proxy for stellar population age (adapted from CFN Chapter 7).

Summary

1. The IMF sets the distribution of stellar masses at birth. Chabrier/Kroupa IMFs predict fewer low-mass stars than Salpeter, leading to lower M_*/L ratios.

2. Star formation histories range from short bursts (small τ) to extended star formation (large τ). The SFH determines how galaxy colors and luminosities evolve.

3. Mass-to-light ratios depend strongly on stellar age and IMF. The M_*/L–color relation allows mass estimation from photometry alone. The Zibetti relation gives $M_* \approx 7.6 \times 10^{10}\,M_\odot$ for the MW.

4. The star formation main sequence shows that SFR scales with M_* for star-forming galaxies, but more massive galaxies have lower specific SFR. There is a clear bimodality between star-forming and quenched populations.